that a policy can control through them, and how much plant variation the resulting closed loop already absorbs unaided. A small answer to RQ6 is then a result about the system rather than a failure of the method, because accurate models and a robust policy leave less for adaptation to recover. The work rests on the following contributions:

Transferable Hydraulic Actuator Learning: A pipeline that identifies hydraulic actuation from high-fidelity Simscape data and transfers it into IsaacLab through two representations, a modular inverse and forward chain and a physics-structured reduced-order model. The inverse model is trained on Simscape data and evaluated on deployment runs drawn entirely outside that training distribution, where it reproduces most of the valve-command variance with an accuracy that is nearly flat across plant domains. It requires no deployment-specific retraining and remains frozen in every downstream stack. The reduced-order model is transferred differently and the difference is stated wherever it matters: it receives a small bounded calibration against the deployment plant, which is why it serves as a reference condition rather than as an independent competitor to the inverse and forward pair.

Control-Relevant Identification and Validation: A demonstration that predictive accuracy can conceal the loss of exactly the property a controller needs, together with the procedure that detects it. The first forward model reached a near-perfect supervised score while learning to ignore the valve command and reconstruct motion from the state history instead. The validation loss did not show this; ablating the command did. Redesigning the excitation and retraining recovered genuine command sensitivity, and the pairing of excitation design with input ablation is established here as a required test for learned hydraulic models intended for control rather than for prediction.

Reduced-Order Actuator Model: A command-conditioned grey-box actuator in which pressure, force, velocity and extension propagation remain explicit while learning covers valve realisation, unmodelled loading and bounded residual effects. It is accurate enough to serve as the deployment plant, stable in recursive rollout, and cheap enough to evaluate inside GPU-parallel policy training. Zeroing or shuffling its command collapses its extension and velocity accuracy, which establishes that the command authority is real rather than inferred from context.

Policy-Only Control Through Learned Hydraulics: A damped least-squares controller supplies privileged demonstrations during training and is then removed, so that the distilled policy emits desired cylinder motion directly into a learned actuator representation and constitutes the entire deployed controller. Both representations support closed-loop terrain following in this configuration, although two learned models placed in series could have compounded their errors.

Robustness and the Limits of Adaptation: A matched evaluation protocol that scores the resulting stacks on tracking, control effort, latency, disturbance rejection and sensitivity to the magnitude of plant variation, together with a procedure that decides in advance whether online plant conditioning will repay the effort. The first of its two measurements establishes whether plant identity is present in the data the machine produces. The second supplies the true plant parameters through a privileged encoder and measures what the closed loop does with them. Both precede estimator training, and either can end the effort before it is incurred.